Hypothesis Testing and A/B Testing

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1. Null and Alternative Hypotheses – Zomato

Scenario:

Zomato wants to know whether a **new restaurant listing design** increases the average number of daily orders compared to the old design.

Let:

- μ_1 = Average daily orders with the new design
- μ_2 = Average daily orders with the old design

Null Hypothesis (H_0):

H_0: $\mu_1 \leq \mu_2$

The new restaurant listing design does **not increase** the average number of daily orders.

Alternative Hypothesis (H_1):

H_1: $\mu_1 > \mu_2$

The new restaurant listing design **increases** the average number of daily orders.

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2. P-value and Null Hypothesis – Flipkart

Given:

P-value = 0.03

Significance level (α) = 5% = 0.05

Decision Rule:

- If **p-value ≤ significance level**, reject the null hypothesis.
- If **p-value > significance level**, do not reject the null hypothesis.

Here:

0.03 < 0.05

Therefore:

Reject the Null Hypothesis (H_0).

Explanation:

Since the p-value of **0.03** is less than the 5% significance level, there is sufficient statistical evidence to conclude that the two versions of the Flipkart checkout page have a significant difference.

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3. A/B Test Plan for Spotify

Objective:

Spotify wants to find out whether a **new playlist recommendation algorithm** increases the daily listening time of users.

Step 1: Divide Users into Two Groups

Group A – Control Group:
Users continue using the existing recommendation algorithm.

Group B – Test Group:
Users receive recommendations from the new algorithm.

Step 2: Hypotheses

Null Hypothesis (H_0):
The new algorithm does not increase average daily listening time.

Alternative Hypothesis (H_1):
The new algorithm increases average daily listening time.

Step 3: Data to Collect

Spotify should collect:

- Average daily listening time per user
- Number of songs played
- Number of playlists listened to
- Number of sessions per day
- User engagement with recommended songs

Step 4: Comparison

After the test period, Spotify compares the **average daily listening time** of Group A and Group B.

Step 5: Decision

If the new algorithm produces a statistically significant increase in listening time, Spotify can conclude that the **new algorithm performs better**.

If there is no statistically significant improvement, Spotify should continue with the existing algorithm or make further improvements.

Main Metric:

Average daily listening time per user (in minutes).

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4. A/B Test for Instagram

Business Question:

Does changing the Instagram Reels recommendation layout increase the amount of time users spend watching Reels?

A/B Test

Group A: Users see the current Reels layout.

Group B: Users see the new Reels layout.

Null Hypothesis (H_0):

The new Reels layout **does not increase** the average time users spend watching Reels.

Alternative Hypothesis (H_1):

The new Reels layout **increases** the average time users spend watching Reels.

Metric to Measure:

Average Reels watch time per user per day (in minutes).

Decision:

If Group B has a statistically significant higher average watch time than Group A, the new layout can be considered **more effective**.